SOFTWARE ENGINEERING
PROJECT PROPOSAL

Drone Defense

I. Project Title

Drone Defense

II. Project Proponent

The project will be proposed and developed by the student project group from the Bachelor of Science in Computer Science – Application Development Track. The group members will be responsible for the planning, requirements gathering, system design, game design, and proposed development of the system.

Project Members: **Balase, Mark Nathaniel C.**
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Gelindon, Justine Clein
Gensaya, Lorens Mark
Grande, Jeremy

Section: **III - CCSAD**

III. Project Description

Our game, “Drone Defense” is a simple wave-based tower defense-like game where you deploy drones to fight for you. It is a strategy and platformer game where you must tactically deploy your allies to the best spots to defend yourself and advance to the next level, leveling up and gaining upgrades as you go on.

IV. The Problem Intended to Be Solved

The game addresses themes of war and psychological warfare. It promotes camaraderie through gameplay and provides a challenge that must be overcome through the power of working side by side with each other.

V. Methodology

The group will use the Agile methodology to develop our game, dividing the process into iterations and delivering updates incrementally. With each iteration, the team will add updates, playtest the game, review bugs, and polish the game further. Furthermore, we will periodically receive feedback from our peers and professors in order to understand the final requirements and features our final game will have.

VI. Techniques Used in Requirements Gathering

We gathered our requirements from consultations with our Elective professor, as well as collective brainstorming sessions.

VII. Target Beneficiaries of the System

The beneficiaries for this project will be the players, or students who will be testing and playing this game.

VIII. Features and Functionalities

a. Main Menu

- An interface where players can get a general idea of the game's aesthetic, as well as containing functionality to proceed to the actual gameplay.

b. Mouse and Keyboard Controls

- WASD to move in a top-down movement system. Using your mouse, you can interact with and move your allies for strategic placement.

c. Wave-Based Enemy Spawning

- This game is played in waves and subwaves, with each subwave spawning a group of enemies which all must be defeated by the player using any tools at their disposal.

d. Dynamic Difficulty / Wave Progression

- Waves of enemies get more difficult as the game progresses, requiring more upgrades and strategy. Subwaves get faster or more difficult based on player performance.

e. Abilities

- Both the player and their allies will have special abilities that will assist in their survival.

f. Ally Control System

- Using the mouse, the player can select an ally to move it anywhere on the field, or to use the ally's abilities manually. They must utilize this system to combat the oncoming onslaught of enemies while keeping the player character safe.

g. Ally Synergy

- Ally can work with one another to combine abilities for a more powerful coordinated attack. This is done by aiming their abilities at other allies for unique interactions.

h. Upgrade System

- After every wave, the player can upgrade either themselves or their drones to increase stats or to unlock abilities as the game goes on.

i. Dialogue

- There will be a simple story and dialogue as waves progress, therefore there will be a system to display character sprites and dialogue to show interaction and communication between characters.

IX. Algorithms

Dynamic Difficulty Adjustment (DDA) & Wave Escalation

Proportional-Integral-Derivative (PID) Controller

If the walls' "survival health" is set to 50%, and the player clears a wave with 100% wall health, the system increases enemy spawn rates for the next subwave. If wall health drops, it will adjust the next subwave spawns slightly and subtly, allowing for a fairer and immersive experience.

Drone Autonomous Targeting & Threat Prioritization

Spatial Partitioning (2D Quadtree / Grid Bucket)

Quadtree Partitioning: Instead of every ally calculating the distance to 50 active enemies every frame, the arena is split into smaller spatial quadrants. Allies only search for targets within their immediate quadrant, keeping frame rates smooth.

Wave Composition & Procedural Squad Generation

Knapsack Optimization (Wave Budget Formulation)

Knapsack Formulation: The wave manager has a "spawning budget" (e.g., Subwave 1 has 20 points). Fast enemies cost 2 points, tanky enemies cost 3 points, and sniper enemies cost 5 points. The knapsack algorithm spends this budget to fill subwaves with balanced combinations that total exactly 100 points.

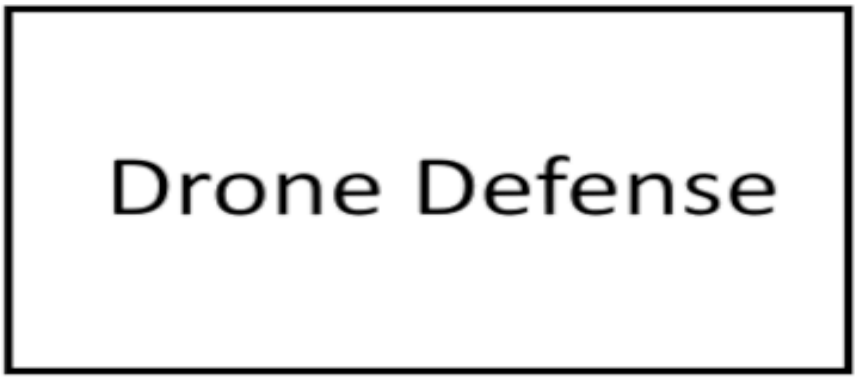
Post-Wave Upgrade System & Build Recommender

Directed Acrylic Graphs (DAGs) using Kahn's Algorithm

Nodes represent upgrades and edges represent requirements such as needing a pre-requisite upgrade or needing to beat a certain level. Kahn's Algorithm helps to disallow and avoid cyclical errors where upgrades may accidentally require themselves to be unlocked.

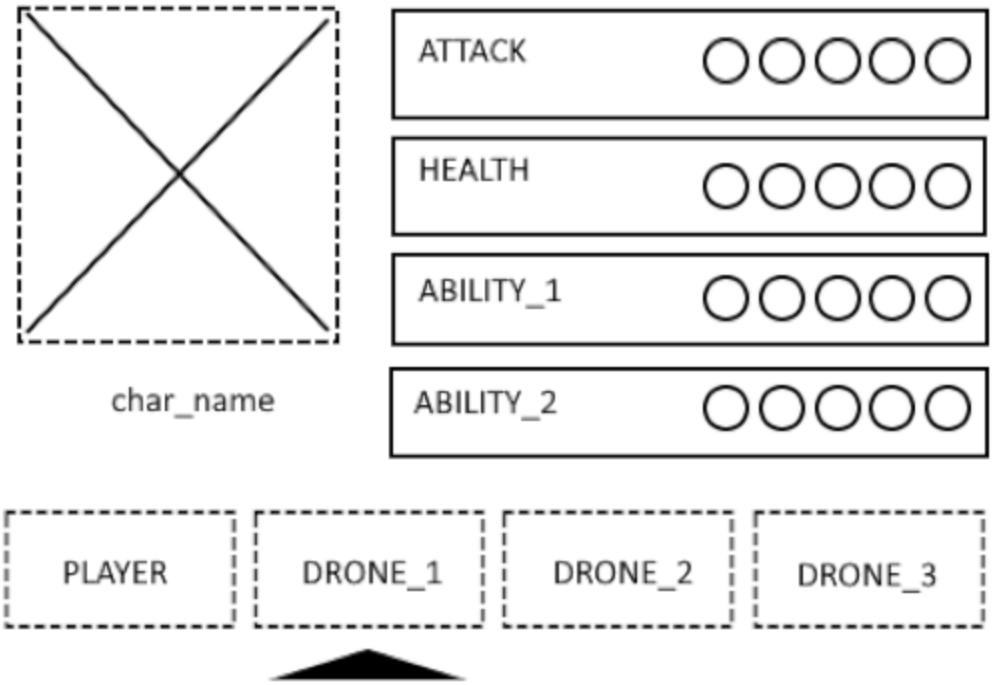
X. Project Prototype

MAIN MENU

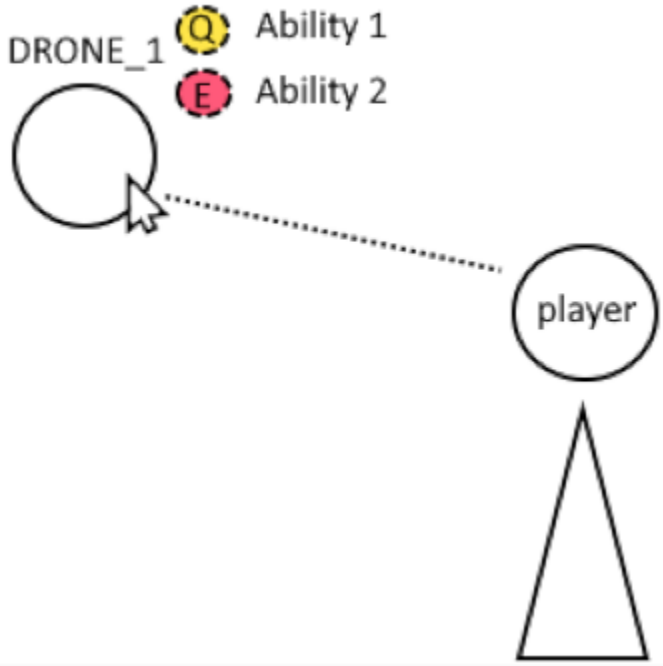




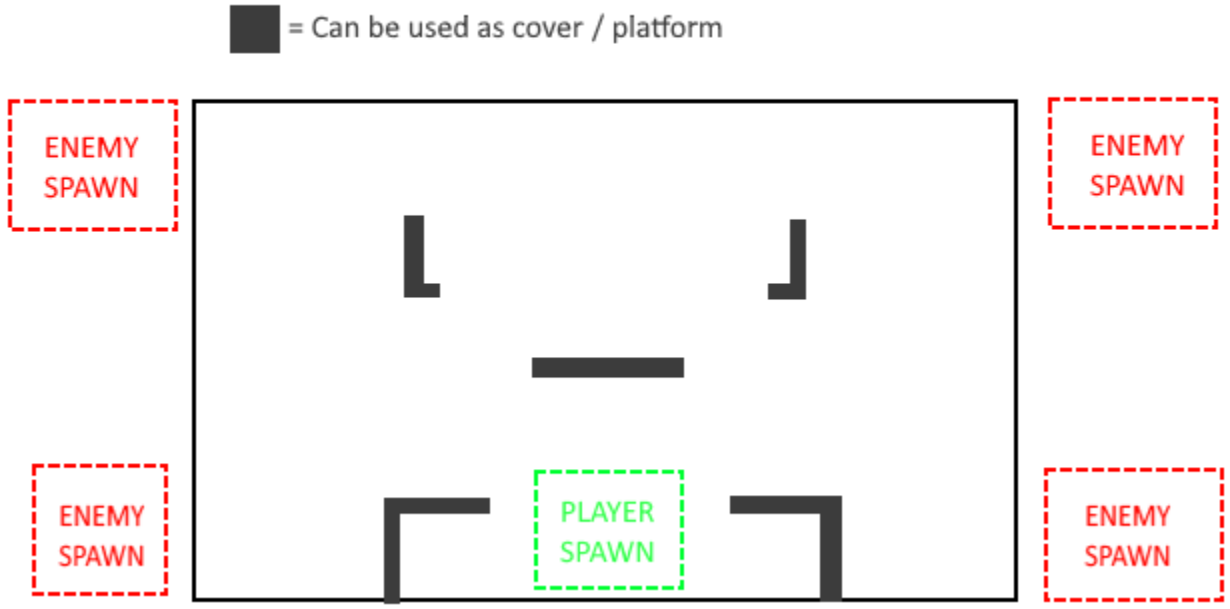
UPGRADE MENU


ATTACK
HEALTH
ABILITY_1
ABILITY_2
char_name
PLAYER
DRONE_1
DRONE_2
DRONE_3

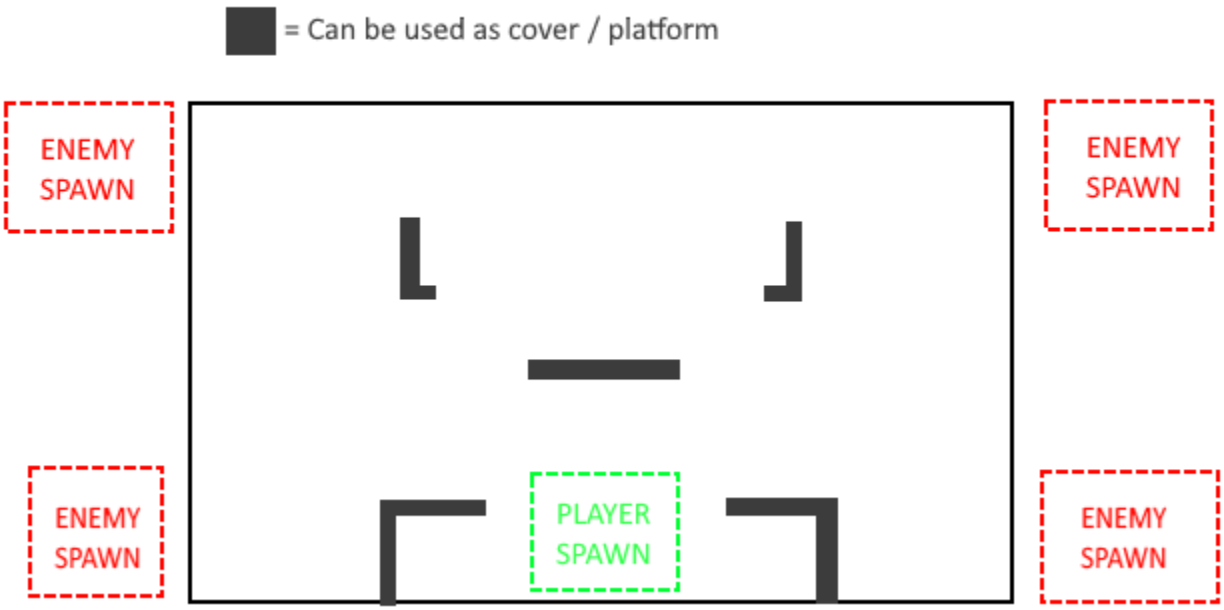
PLAYER / DRONE SELECT



MAP 1



MAP 2


= Can be used as cover / platform
ENEMY SPAWN
ENEMY SPAWN
ENEMY SPAWN
PLAYER SPAWN
ENEMY SPAWN